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Influence of electrolyte conductivity on the performance of lithium-ion batteries: An electrochemical impedance analysis using symmetric cells

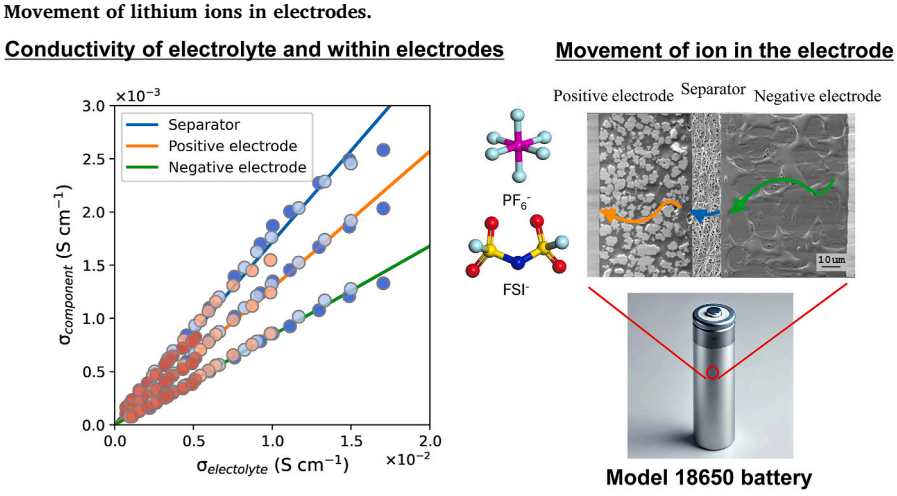
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HIGHLIGHTS

- The quantitative influence of the electrolyte conductivity on battery is determined.
- The tortuosity factors of the electrodes were calculated by electrochemical analysis.
- The tortuosity factor significantly influences battery performance.
- The LiFSI electrolyte has high conductivity and improves battery performance.

GRAPHICAL ABSTRACT



ARTICLE INFO

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ABSTRACT

Lithium-ion batteries are widely used in energy storage applications owing to their high energy density, long cycle life, and efficiency. The electrochemical performance of cylindrical lithium-ion batteries is significantly influenced by the choice of electrolytes. This study examines the impact of four different electrolytes-lithium bis (fluor sulfonyl) imide (LiFSI), lithium hexafluorophosphate ($LiPF_6$), lithium bis (pentafluoroethanesulfonyl) imide (LiBETI), and lithium tetrafluoroborate ($LiBF_4$)-on the battery performance by evaluating their influence on the conductivity, resistance components, and overall electrochemical behavior. The ionic conductivity of the electrolytes, measured using evaluation cells, is significantly influenced by the anionic species, with LiFSI exhibiting the highest conductivity among the tested electrolytes. Additionally, electrochemical impedance spectroscopy was performed using symmetric cells to determine the contributions of various resistance components. The results indicate that LiFSI exhibited the highest conductivity and lowest resistance, followed by $LiPF_6$ and LiBETI, with $LiBF_4$ exhibiting the poorest performance. The relationship between the ionic conductivity within the electrode and the ionic conductivity of the electrolytes are measured over a wide temperature range, and the tortuosity factors of the electrodes are evaluated. The relationship between the ionic conductivity

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of the electrolyte and the ionic conductivity in the electrode is consistent across different electrolytes. Furthermore, the ionic resistance in the electrode is determined using the ionic conductivity of the electrolyte and the tortuosity factor of the electrode. Among the studied electrolytes, LiFSI demonstrated the highest ionic conductivity, leading to reduced ionic resistance in the electrode and improved battery performance.

1. Introduction

Lithium-ion rechargeable batteries are commonly used in electronic and electrical devices, including smartphones and portable computers. More recently, electric vehicles, hybrid electric vehicles (HEVs), plug-in hybrid electric vehicles (PHEVs), and railways have adopted such batteries [1–3]. The key performance parameters of lithium-ion rechargeable batteries for automotive applications include energy density, power capability, and cycle life. Improving these parameters requires a thorough understanding of the electrochemical reaction mechanisms of porous electrodes. Several studies have modeled the charge-discharge behavior of porous electrodes, including those of lithium batteries, based on Newman's famous electrochemical reactions and ionic transport equations [4–8]. However, these models require precise knowledge of the physical properties of the active material and electrolyte, along with the electrochemical parameters of the electrode. To address this, we previously developed an electrochemical method based on electrochemical impedance spectroscopy using symmetric cells (EIS-SC) for the analysis of porous electrodes based on a transmission line model (TLM), which enabled the internal resistance components of electrodes to be evaluated. In particular, we analyzed the electrolyte resistance between opposing electrodes (R_{sol}), ionic resistance within the electrode pores (R_{ion}), and charge-transfer resistance for lithium intercalation (R_{ct}), as illustrated in Fig. 1 [9]. Further analysis of the correlation between electrode parameters, such as thickness and conductive carbon ratio, and their impact on performance showed that ion resistance increases proportionally with electrode thickness, resulting in thick electrodes designed for high-energy-density batteries having lower power capability: however, the performance can be enhanced by reducing the tortuosity factor and carbon ratio to optimize the electrode configuration [10,11]. Several studies have explored both modeling and experimental approaches to reduce tortuosity and enhance the performance of thick-film electrodes[12–16].

Our previous evaluation of the durability of cylindrical batteries at elevated temperatures using $LiNi_xCo_yAl_zO_2$ (NCA)- and $LiNi_xCo_yMn_zO_2$ (NCM)-based positive electrode materials identified that degradation at the interface between the positive active material and electrolyte resulted in the loss of both capacity and power [17,18]: however, elemental substitution of the positive active material could inhibit this degradation [19,20]. These studies primarily used lithium hexafluorophosphate ($LiPF_6$) as the electrolyte: thus, the impact of the properties of the electrolyte on the resistance components remains unclear. For instance, while ionic resistance in the electrode is known to be influenced by the electrolyte conductivity and electrode tortuosity, the quantitative effect of the electrolyte conductivity on the ionic resistance has not been fully investigated [21,22]. $LiPF_6$ remains the most common electrolyte in lithium-ion batteries: however, high-conductivity alternatives such as Lithium Bis (fluor sulfonyl) imide (LiFSI) have recently been developed and commercialized [23,24]. Further improving the performance of lithium-ion batteries requires a thorough understanding of the influence of the physical properties of the electrolyte on the resistance components. Additionally, the electrolyte composition affects electrode interface degradation at high temperatures and is therefore expected to influence the long-term stability of the battery.

While $LiPF_6$ remains the most commonly used salt in commercial lithium-ion batteries due to its favorable balance of ionic conductivity

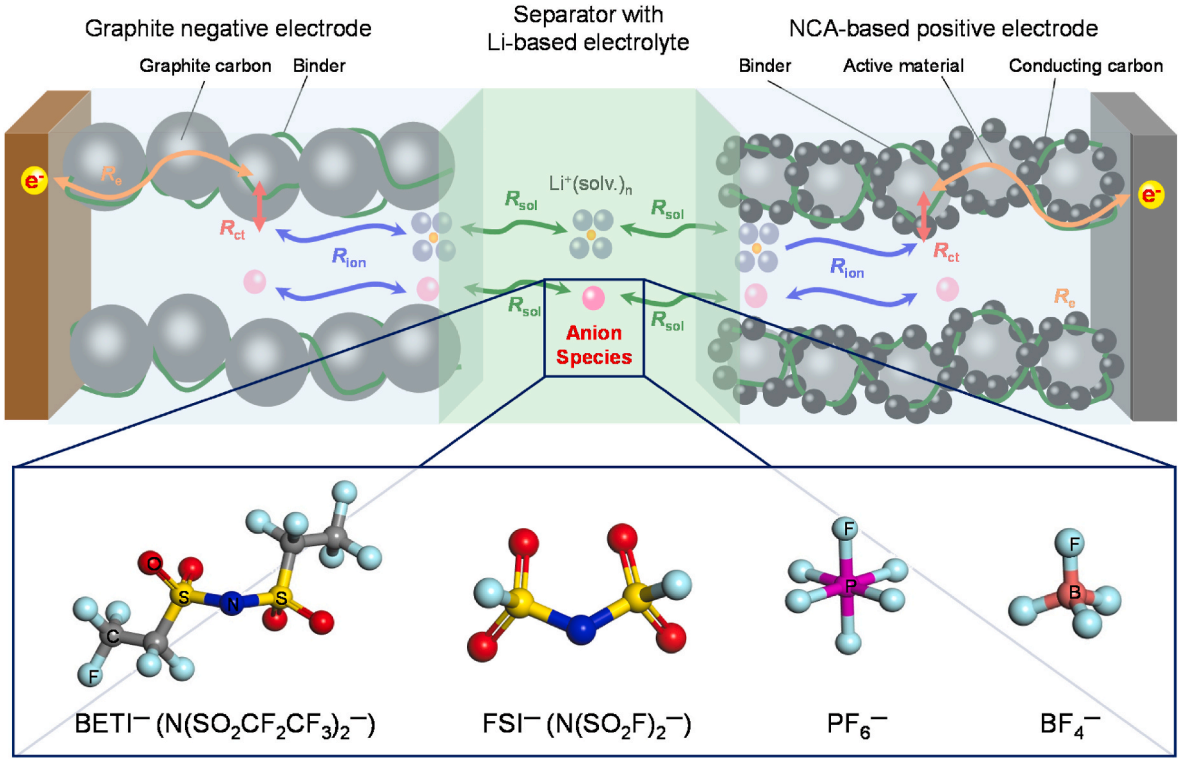


Fig. 1. Schematic illustration of the internal resistance components within porous electrodes and molecular structures of $BETI^-$, FSI^-, PF_6^-, and BF_4^- anions in lithium-ion batteries.

and interfacial stability, alternative lithium salts have gained attention for their unique properties. LiFSI exhibits higher ionic conductivity and improved solid electrolyte layer (SEI) formation behavior, while mitigating issues such as aluminum current collector corrosion that are known to affect lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) at high potentials. Lithium bis(pentafluoroethanesulfonyl)imide (LiBETI), with its large and hydrophobic anion, offers promising oxidative stability, albeit with somewhat lower dissociation efficiency. LiTFSI has been widely studied as a high-conductivity salt, it suffers from severe drawbacks such as aluminum corrosion and unstable SEI formation, limiting its practical use. LiFSI and LiBETI have emerged as improved alternatives, offering better anodic stability and interfacial behavior, making them more suitable for practical battery applications [21,24,27]. Lithium tetrafluoroborate ($LiBF_4$), although historically used, tends to form less stable SEI layers and has limited dissociation behavior. These salts differ in their anionic size, structure, and ion-association tendencies, which may influence not only bulk conductivity but also ionic transport within porous electrodes. We systematically compare four electrolyte salts—$LiPF_6$, LiFSI, LiBETI, and $LiBF_4$—dissolved in carbonate-based solvent, to investigate how their ionic properties impact electrode-level resistance and battery performance.

In this study, we developed a prototype cylindrical cell battery [25] to demonstrate the performance [23] of four types of electrolytes with distinct physical and electrochemical properties. The initial performance (capacity and power capability) and high-temperature cycle durability of the electrolytes were assessed to determine their practicality. Furthermore, electrochemical analysis using symmetric cells was employed to quantify the resistance components of the porous electrodes. Analysis of the relationship between the electrolyte conductivity and ionic resistance of the electrode over a wide temperature range, we quantitatively determined the impact of electrolyte properties on the resistance and power capability of the battery was quantitatively determined by examining the relationship between the electrolyte conductivity and ionic resistance of the electrode over a wide temperature range.

2. Theory and experimental methods

2.1. Impedance theory for cylindrical pores using transmission line model (TLM) [9,26]

In TLM-based analysis, the total impedance contributions from non-faradaic (Z_{nf}) and faradaic (Z_f) procedures can be expressed by equations (1) and (2):

$$Z_{nf,\omega} = R_{sol,L} + \sqrt{\frac{R_{ion,L}}{j\omega C_{dl,A} 2\pi r}} \coth \sqrt{R_{ion,L} j\omega C_{dl,A} 2\pi r} L, \tag{1}$$

$$Z_{f,\omega} = R_{sol,L} + \sqrt{\frac{R_{ion,L} R_{ct,A}}{(1 + j\omega R_{ct,A} C_{dl,A}) 2\pi r}} \coth \sqrt{\frac{R_{ion,L}(1 + j\omega R_{ct,A} C_{dl,A}) 2\pi r}{R_{ct,A}}} L, \tag{2}$$

Here, $R_{ion,L}$ is the ionic resistance per unit pore length (Ω/cm), which is given by:

$$R_{ion,L} = \frac{\rho}{\pi r^2}, \tag{3}$$

where $R_{sol,l}$ is the electrolyte resistance per unit length in the separator between two electrodes, and R_{sol} is the electrolyte resistance in the separator between two electrodes with respect to l as:

$$R_{sol} = R_{sol,l} \cdot l, \tag{4}$$

The ionic resistance in the pores (R_{ion}) with respect to L is expressed as:

$$R_{ion} = R_{ion,L} \cdot L, \tag{5}$$

Similarly, the total charge transfer resistance (R_{ct}) with respect to L is given by:

$$R_{ct} = \frac{R_{ct,A}}{2\pi r L}, \tag{6}$$

where $R_{ct,A}$ is the charge transfer resistance per unit of electroactive surface area (Ω cm^2).

The limiting values of the real parts (as $\omega \to 0$) in the non-faradaic ($Z'_{nf,\omega \to 0}$) and faradaic ($Z'_{f,\omega \to 0}$) processes are given by Eqs. (7) and (8), respectively [26].

$$Z'_{nf,\omega \to 0} = \frac{R_{ion}}{3}, \tag{7}$$

$$Z'_{f,\omega \to 0} = \frac{R_{ion}}{3} + R_{ct}, \tag{8}$$

This combined approach enables the separation and quantification of individual electrochemical parameters, providing a detailed understanding of resistance contributions within porous electrodes [9,10].

2.2. Materials

2.2.1. Electrolytes

The electrolytes used were 1.0 M solutions of $LiPF_6$, $LiBF_4$, LiBETI, or LiFSI, each dissolved in a 30:70 vol ratio of ethylene carbonate (EC) and ethyl methyl carbonate (EMC). All electrolytes used in this study were custom prepared by Kishida Chemical Co., Ltd. (Japan) according to our specified compositions. The solutions were of Lithium Battery Grade, with component purities ≥99.5 %. Moisture and HF contents were ≤20 ppm and ≤50 ppm, respectively, and trace metal impurities (Cl^-, Na^+, K^+, Ca^{2+}, Fe) were below 5 ppm or less. The electrolytes were synthesized in a highly controlled glovebox environment as assured by the supplier's certification. Fig. 1 shows the molecular structure of each electrolyte. Although $LiPF_6$ is the most common electrolyte used in lithium-ion batteries, LiFSI has recently garnered significant interest owing to advancements in industrial-scale mass production [24]. Additionally, the ionic radii of the anions in these electrolytes follow the trend $BF_4^- < PF_6^- < FSI^- < BETI^-$: thus, this study also aimed to determine the influence of the ionic radii on interactions of the anion with the porous structure of the electrode [28,29].

Cost-wise, $LiPF_6$ remains the reference salt at approximately 8 USD/kg in large-scale production, while LiFSI currently costs around 60–250 USD/kg—1–2 orders of magnitude higher—but shows significant adoption growth and expected cost reduction with scale. $LiBF_4$ is similarly priced to $LiPF_6$, whereas LiBETI remains niche and comparatively expensive.

2.2.2. Electrolyte conductivity measurement

A sandwich-type cell with an electrolyte between two Pt electrodes was used as the evaluation cell. The cell constant was pre-determined using a standard aqueous KCl solution for conductivity measurements, with a conductivity of 12.88 mS/cm at 25 °C. AC impedance measurements (frequency range: 1 Hz-100 kHz, amplitude 100 mV) were performed at temperatures ranging from −30 °C to 60 °C. (Specifically, it is measured at temperatures of −30, −20, −10, 0, 10, 20, 25, 30, 40, 50, and 60 °C.)

2.2.3. Electrode material

The NCA-based positive active material incorporated carbon black and polyvinylidene fluoride as a conductive additive and binder, respectively. $LiNi_{0.75}Co_{0.15}Al_{0.05}Mg_{0.05}O_2$ was synthesized as the positive active material using the coprecipitation method [19]. The positive active material, conductive carbon, and binder were mixed in a ratio of 85:10:5 (wt.%). The positive electrode mixture was dispersed in

N-methyl-2-pyrrolidone to prepare a paste that was spread on the surface of 15 μm thick aluminum foil and allowed to dry. The typical coat loading per unit area (one side) of the positive electrode composite was 10.5 mg/cm^2, and the positive electrode was obtained by adjusting the density to 2.3 g/cm^3 (porosity: 0.38) using a roll press. To prepare the carbon-based negative electrodes, a composite of artificial graphite and binder (95:5 wt%) was spread on surface of a 10 μm thick copper foil. The typical loading per unit area (one side) of the negative electrode composite was 7.5 mg/cm^2, and the density was adjusted to 1.2 g/cm^3 (porosity: 0.42) using a roll press to obtain the negative electrode.

2.3. Electrochemical measuring cell, electrochemical measurement

2.3.1. Fabrication and evaluation of cylindrical type batteries

Positive and negative electrode sheets measuring 54 mm × 450 mm and 56 mm × 500 mm, respectively, were rolled together with a separator consisting of a microporous polyethylene membrane (porosity: 0.40), to form the cylindrical battery [25]. The batteries also included the four electrolytes described in section 2.2.1. We measured the weight difference before and after electrolyte injections for all small cells assembled. From this, we calculated the actual volume of electrolyte injected and compared it with the theoretical pore volume of the positive and negative electrodes and the separator. As shown in Fig. S1 (Supplementary Information), all cells exhibited an injected electrolyte amount between 100 % and 120 % of the calculated pore volume, confirming that complete wetting was achieved. The small excess (>100 %) likely corresponds to residual electrolyte between the electrode and the inner wall of the casing. The batteries were charged to 4.1 V with a constant current of 0.1 mA/cm^2 and constant voltage, then discharged to 3.0 V under the same conditions at 20 °C. The first discharge capacity was considered the initial capacity of each battery. The direct current (DC) resistance (R_{IV}) of the batteries was calculated from the shift in the electric potential (ΔV) upon application of DC (I) to the battery for 10 s at a state of charge (SOC) of 50 %, using equation (9):

$$R_{IV} = \frac{\Delta V}{I}, \tag{9}$$

The power capability (W) was calculated from SOC 50 % using equation (9):

$$W = \frac{(V_{SOC50\%} - V_{LOP})}{R_{IV}}, \tag{10}$$

where $V_{SOC50\ \%}$ is the battery voltage at SOC 50 % and V_{LOP} is the lowest operating potential (3.0 V at 20 °C and 2.5 V at −30 °C).

The cycle test was performed in the voltage range of 3.0 V–4.1 V at a constant current density of 2 mA/cm^2 (corresponding to a 2C rate). The initial capacity and DC resistance of each battery were compared with those measured after 500 cycles under the same conditions.

2.3.2. Electrochemical impedance analysis using symmetric cells [9]

The resistance components of the porous electrodes were determined by applying the theory of TLM and EIS–SC. The analysis non-faradaic processes by EIS-SC was performed to determine the R_{ion} of the positive uncharged electrode. Two sheets of uncharged positive electrodes were opposed by a separator to prevent electrical transfer. The R_{ion} of the negative electrode was determined using the same approach. To measure the R_{ct}, two sets of cells with positive and negative electrodes were opposed by a separator and operated for a minimum of one cycle in the range of 3.0 V–4.1 V at a constant current density of 0.1 mA/cm^2 and charged at 3.685 V to set SOC50 %. Subsequently, the positive and negative electrodes were removed, and symmetric cells consisting of both charged positive electrodes and both charged negative electrodes were reassembled in a glove box, respectively. AC impedance spectroscopy was performed using a Solartron 1260/1286 in the frequency range of 0.01 Hz–100 kHz with a voltage amplitude of 10 mV. Measurements were conducted between −30 °C and 60 °C. The performances of the symmetric cell and battery were then compared. In this study, as mentioned above, impedance spectroscopy was conducted at SOC 0 % and 50 % to enable standardized and meaningful comparisons across electrolytes. These SOC levels were selected to represent typical operating conditions in HEV/PHEV applications and to minimize confounding effects from phase transitions or interfacial reactions that are highly SOC-dependent.

3. Results and discussion

3.1. Initial capacity and power capability of batteries

The initial charging capacity of each battery was 185 mAh/g, while the discharge capacity was 158 mAh/g. These values remained nearly identical across the different electrolytes. Fig. 2 (a) shows the DC resistance values of batteries using various electrolytes at 20 °C. The figure shows the average value of the five cells and the standard error as an error bar, and the error is small enough to show that a significant difference is observed in each electrolyte (Fig. 2 (a) also shows a schematic illustration of a cylindrical type battery.). Among them, batteries with the LiFSI electrolyte exhibited the lowest resistance, followed by those with $LiPF_6$ and LiBETI, while batteries using the $LiBF_4$ electrolyte had the highest resistance. Fig. 2 (b) illustrates the power capability of cylindrical batteries with each electrolyte at 20 °C and −30 °C (with five data points obtained under each condition). The power capability of the batteries was higher when the DC resistance was lower - batteries using the LiFSI electrolyte had the highest power capability, followed by those with $LiPF_6$ and LiBETI, whereas batteries with the $LiBF_4$ electrolyte had the lowest output.

3.2. Cycle characteristics at elevated temperature

The 60 °C cycling test was chosen as it reflects a realistic thermal stress condition for automotive applications. Previous studies [25] have shown that while room temperature cycling yields relatively stable performance, battery degradation was substantially more pronounced at 60 °C. Therefore, we selected 60 °C as a meaningful condition to highlight durability differences among electrolytes in a high-stress operating window relevant to automotive applications.

A preliminary durability test was conducted on two cells per electrolyte type at 60 °C. While the sample size was limited, the observed differences in capacity retention and resistance increase were consistent as described below. These results are presented in the Supporting Information for reference. Figure S2 (a) illustrates the cycle performance of these batteries at 60 °C, demonstrating the change in the capacity retention rate over 500 cycles. (For details of the charge-discharge curve, see Fig. S3. The battery with the LiFSI electrolyte exhibited the highest capacity retention rate, followed by those with $LiPF_6$ and LiBETI, while the battery with the $LiBF_4$ electrolyte had the lowest capacity retention rate. Figure S2 (b) illustrates the initial capacity, capacity after 500 cycles (for details of the charge/discharge curves before and after the cycle test, see Fig. S4., and capacity recovery rate, which is defined as follows:

$$\textit{Capacity retention rate} = \frac{\textit{capacity after 500 cycles}}{\textit{initial capacity}} \times 100. \tag{11}$$

The differences in retention shown in Figures S2 (a) and S2 (b) arise from the measurement protocols. Figure S2 (a) reflects in situ cycling behavior at 60 °C(CC/CC), while Figure S2 (b) compares battery capacities measured under CCCV conditions before and after 500 cycles at 20 °C. As such, Figure S2 (b) excludes resistance effects, providing a clearer view of intrinsic capacity loss. The capacity retention rate followed the order: LiFSI > $LiPF_6$ ≥ LiBETI > $LiBF_4$. The differences in capacity retention rates were attributed to the following factors: While detailed interfacial analysis was not performed in this study, previous

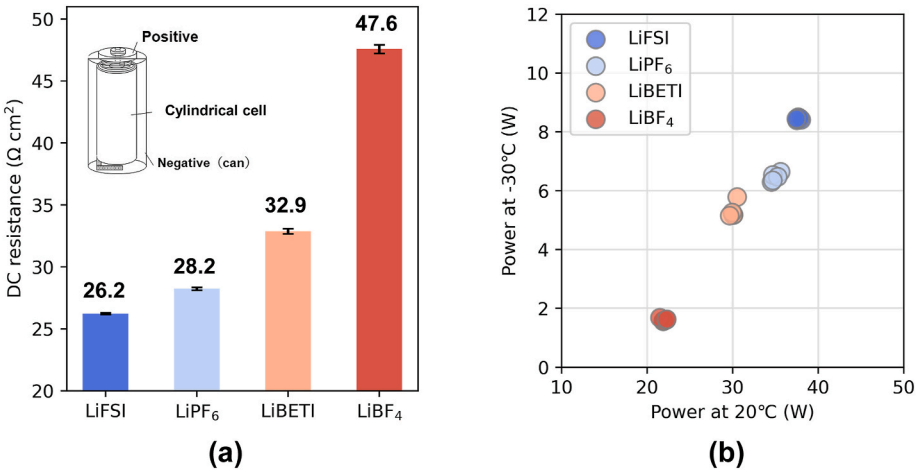


Fig. 2. (a) DC resistance of the batteries using various electrolytes at 20 °C (average and standard error of five data points). Inset: Schematic diagram of a cylindrical battery. (b) The power capability of cylindrical batteries with each electrolyte at 20 °C and −30 °C (five data points each).

reports have suggested that LiFSI tends to form a more stable SEI films enriched in LiF than $LiPF_6$, which may help explain the observed durability differences among electrolytes [31]. In contrast, SEI formed with $LiBF_4$-based electrolytes has been reported to be less stable [30]. LiBETI, while offering good thermal and oxidative stability, may form thicker interfacial layers with limited ion transport. These effects contribute to the differences in cycling behavior observed in this work. Figure S2 (c) compares the DC resistances of the batteries with before and after 500 cycles. Batteries with $LiBF_4$ and LiBETI electrolytes exhibited a significant increase in resistance after cycling, whereas the increase in the resistance of LiFSI was comparable to that of $LiPF_6$. (Although the capacity retention values for $LiPF_6$ and LiBETI were similar after 500 cycles, the increase in DC resistance was more effectively suppressed in the $LiPF_6$ system. Therefore, $LiPF_6$ is considered to offer better overall durability.) In NCA-based batteries using $LiPF_6$ electrolytes, surface degradation of the positive electrode active material in contact with the electrolyte is known to be a key factor in the increased resistance of the battery. Because the rate of resistance increases for LiFSI and $LiPF_6$ was similar, it is presumed that their degradation mechanisms are not significantly different: however, further investigation is required for a detailed analysis. All electrochemical testing was conducted within a voltage range of 3.0–4.1 V. Previous studies have shown that aluminum corrosion typically occurs at voltages above 4.3–4.5 V, particularly in electrolytes with TFSI-type anions. LiFSI and LiBETI have demonstrated anodic stability up to 4.2 V [21,29], suggesting that Al corrosion is unlikely under our test conditions. This interpretation is further supported by the absence of any abnormal resistance increase attributable to collector degradation.

In summary, LiFSI not only provides superior initial performance but also exhibits greater durability than $LiPF_6$, making it a promising electrolyte for future applications. These differences in long-term stability may be associated with interfacial reactions and SEI/cathode electrolyte interface (CEI) formation, which are known to be influenced by electrolyte anion species. While direct interfacial analysis was beyond the scope of this study, literature reports support this interpretation [30,31]. Table S1 shows a summary of representative literature on SEI and CEI formation for different electrolyte anions. These references support the interpretation of durability differences discussed in this section.

In addition, a single-cell cycle test at 0 °C was conducted to examine low-temperature durability. Although limited in sample size, the capacity retention trend again followed the order LiFSI > $LiPF_6$ > LiBETI > $LiBF_4$. These results are shown in Supporting Fig. S5 and support the view that high ionic conductivity helps suppress non-uniform lithium-ion transport during low-temperature operation.

3.3. Electrode resistance component analysis

As discussed earlier, the DC resistance of batteries varies significantly depending on the type of electrolyte, with LiFSI electrolytes exhibiting lower resistance than the other electrolytes. To investigate the underlying reasons for these differences, a resistance component analysis was conducted EIS-SC [9]. Fig. 3 shows the impedance spectra of the symmetric cells for the positive electrodes using different electrolytes, while Fig. 4 shows the results for the negative electrodes. The blue lines represent the impedance spectra of the non-faradaic processes using untreated electrodes, whereas the purple lines correspond to the impedance spectra of the faradaic processes using charged electrodes (SOC 50 % setting) for (a) LiFSI, (b) $LiPF_6$, (c) LiBETI, and (d) $LiBF_4$. Based on these results, the values of the various resistance components (R_{sol}, R_{ion}, and R_{ct}) were in the order $LiBF_4$ > LiBETI > $LiPF_6$ > LiFSI. The equivalent circuit used for impedance fitting show Fig. S6(a)–S6(b). Fig. S6(a) shows the circuit used at SOC 0 % (uncharged), and Fig. S6(b) shows the circuit at SOC 50 %. These have been added to the Supporting Information. The fitted values of R_{sol}, R_{ion}, and R_{ct} for each electrolyte are summarized in Table S2 in the Supporting Information. These values complement the trends shown in Figs. 3–5 and were extracted using ZView fitting procedures with fitting errors below ±2 %. As an example, line diagrams fitted to Fig. 3 (a) are shown in Fig. S7.

To quantify these resistance components, curve fitting was applied to the impedance spectra in Figs. 3 and 4, extracting values for R_{sol}, R_{ion}, and R_{ct}. Fig. 5 (a) illustrates the sum of the R_{sol} and R_{ion} of the positive and negative electrodes, along with the R_{ct} of both electrodes, in each electrolyte. The total resistance followed the same trend observed in the DC resistance values of cylindrical batteries illustrated in Fig. 2 (a), with LiFSI exhibiting the lowest total resistance, followed by $LiPF_6$, LiBETI, and $LiBF_4$. Further analysis of the resistance breakdown revealed that R_{sol} and R_{ion}, collectively referred to as region 1, have a greater impact on battery performance than R_{ct}, referred to as region 2. This suggests that improving electrolyte conductivity and ion transport properties more effectively reduces battery resistance than optimizing charge transfer kinetics alone.

The relationship between the resistance values obtained from the impedance analysis of the cell and the DC resistance of the battery measured over 10 s was examined [32]. Fig. 5 (b) illustrates the relationship between the DC resistance of the cylindrical batteries and the total resistance of the symmetric cells with positive and negative electrodes for each electrolyte. The black dotted line in Fig. 5 (b) represents the equation y = x, where data points would align if the DC resistance and the symmetric cell resistance were identical: however, the observed DC resistance over 10 s is consistently higher than the resistance

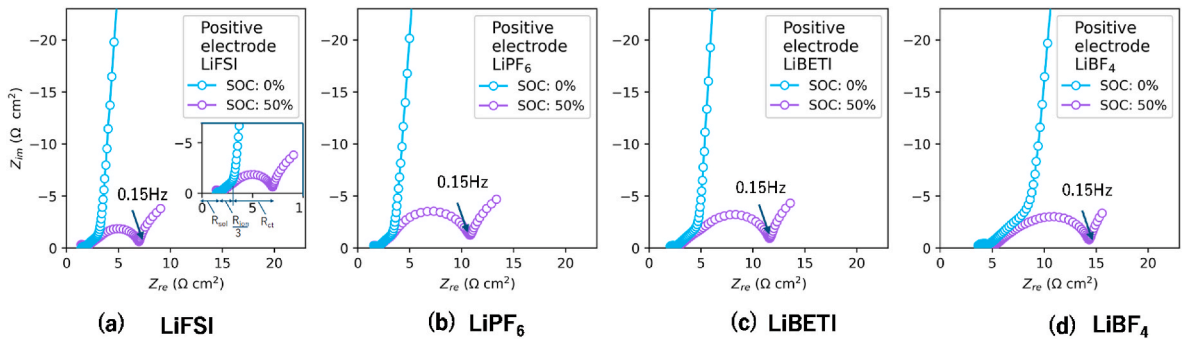


Fig. 3. Nyquist plots of symmetric cells with two positive electrodes in 1.0 M of each electrolyte in EC/EMC (30/70 vol ratio) at 20 °C. (a) LiFSI, (b) $LiPF_6$, (c) LiBETI, and (d) $LiBF_4$.

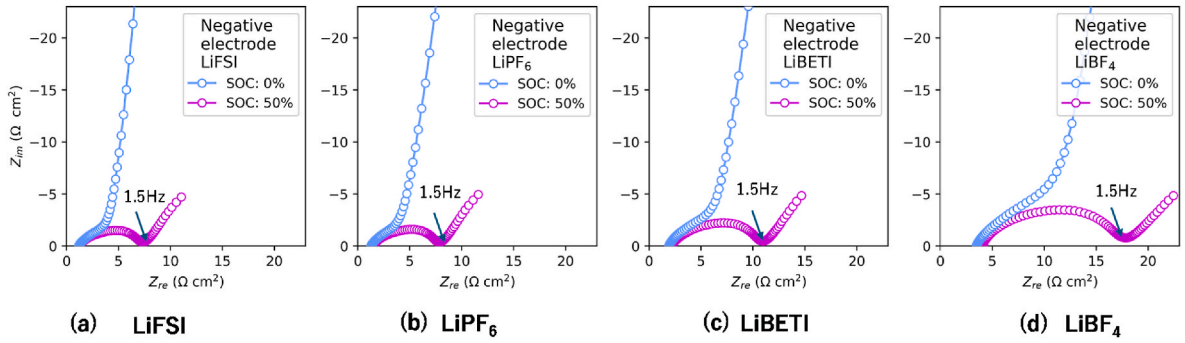


Fig. 4. Nyquist plots of symmetric cells with two negative electrodes in 1.0 M of each electrolyte in EC/EMC (30/70 vol ratio) at 20 °C. (a) LiFSI, (b) $LiPF_6$, (c) LiBETI, and (d) $LiBF_4$.

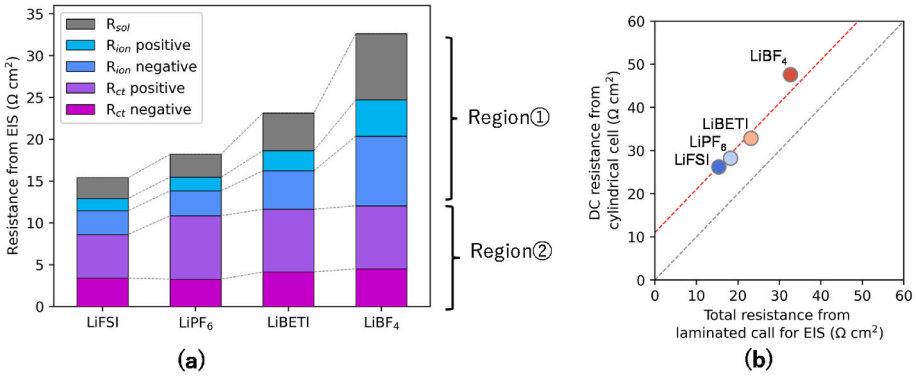


Fig. 5. (a) The sum of resistance components of the cells with positive and negative electrodes. (b) The relationship between the DC resistance of cylindrical batteries and the sum of the resistances of symmetric cells with positive and negative electrodes.

measured in symmetric cells using alternating current. This relationship is represented by the red dotted line y = x + b, where the intercept b corresponds to additional resistance contributions. This intercept resistance **b** is primarily attributed to diffusion resistance, which arises from time-dependent transport processes that occur after electrochemical reactions at the positive and negative electrodes (Figs. 3 and 4). Notably, the value of **b** does not exhibit significant variation across different electrolytes, suggesting that diffusion resistance remains relatively constant regardless of the electrolyte type.

As discussed earlier, R_{sol} and R_{ion} are key factors influencing the power capability of the battery. Therefore, the next section explores the relationship between the physical properties of the electrolyte (particularly ionic conductivity) and resistance. While previous studies have examined the effects of temperature and tortuosity factor on the negative electrode of $LiPF_6$-based batteries [33], optimizing battery performance also requires consideration of the positive electrode and separator, especially when using higher-conductivity electrolytes.

3.4. Conductivity of the electrolyte and conductivity in the electrode

Fig. 6 (a) illustrates the conductivity of various electrolytes measured between −30 °C and 60 °C, plotted as a function of 1000/T (1/K). Across the entire temperature range, LiFSI exhibited the highest conductivity, followed by $LiPF_6$ and LiBETI, while $LiBF_4$ had the lowest. Fig. 6 (b) illustrates the conductivity of electrolytes between the electrodes in the separators, while Fig. 6 (c) and 6 (d) illustrate the conductivity of electrolytes in the positive and negative electrodes, respectively, measured over a temperature range of −30 °C–60 °C using symmetric cells. The equivalent circuit used for impedance fitting, initial parameters, and fitting error percentages are provided in the Supporting

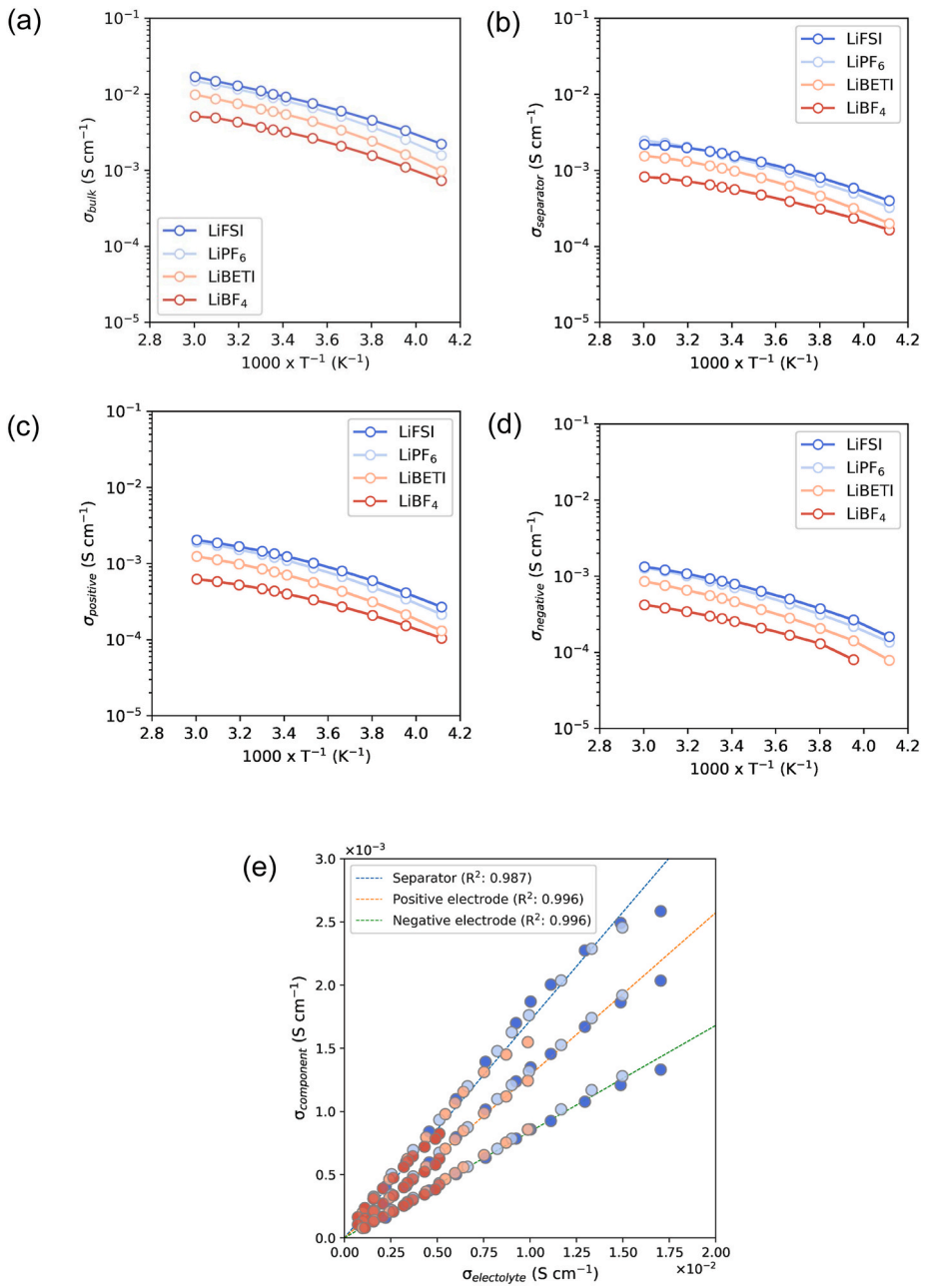


Fig. 6. (a) Ionic conductivity of each electrolyte, (b) Ionic conductivity of each electrolyte between the electrodes in the separator. Ionic conductivity of the electrolyte in the (c) positive and (d) negative electrodes. (e) The relationship between the bulk electrolyte conductivity and that of the electrolyte within the electrode.

Information (Fig. S6, Table S3 and Fig. S8). The typical fitting error was <2 %, ensuring reliable extraction of electrochemical parameters. The trend in conductivity remained consistent across all components: LiFSI > $LiPF_6$ > LiBETI > $LiBF_4$.

The variation in R_{ion} observed among the electrolytes can be attributed not only to conductivity but also to differences in the degree of ion dissociation. Electrolytes with more complete dissociation provide a higher concentration of mobile ions, leading to lower R_{ion}. Based on literature reports and NMR diffusion data, the relative degree of dissociation follows the trend: LiFSI > $LiPF_6$ ≥ LiBETI > $LiBF_4$ [21,24,28]. This trend aligns well with the experimentally observed ionic conductivities and extracted resistance values.

Based on these results, Fig. 6 (e) illustrates the relationship between the conductivity of the bulk electrolyte and that within the electrodes and separators. The observed trends were similar across different components (positive electrode, negative electrode, and separator), regardless of the electrolyte type. The tortuosity factor (τ) of each component is given by:

$$\tau = \frac{\sigma_{electrolyte}\varepsilon}{\sigma_{ion}}, \tag{12}$$

where $\sigma_{electrolyte}$ is the bulk electrolyte conductivity, σ_{ion} is the electrolyte conductivity within the electrodes or separator, and ε is the porosity of the electrode and separator. In Fig. 6 (e), the $\sigma_{electrolyte}$ obtained from Fig. 6 (a) are plotted against the corresponding effective σ_{ion} extracted from Fig. 6(b)–(d). Each line represents a porous medium (separator, positive or negative electrode), and the linear trends confirm the consistent derivation of tortuosity across all electrolyte systems. Tortuosity values were determined based on experimentally measured ionic conductivities both $\sigma_{electrolyte}$ and σ_{ion}. Since EIS was conducted under

low-amplitude AC conditions, concentration polarization effects are negligible. Tortuosity values were only determined for uncharged electrodes. Charged-state conditions were evaluated as the value of R measurements instead, due to structural complexity. The dotted lines in Fig. 6 (e) represent the linear approximation of each relationship, with correlation coefficients indicating a strong positive correlation between the bulk electrolyte conductivity and that within the electrodes.

Based on the aforementioned results, the relationship between the temperature and tortuosity factor of the separator and two electrodes is illustrated in Fig. 7. The dotted lines represent the tortuosity factor, calculated from the slope obtained through linear approximation in Fig. 6 (e), and the porosity. These results indicate that the tortuosity factor depends on the conductivity of the electrolyte, regardless of its type, and remains relatively constant within the variation range for each electrode. Although electrolyte swelling and thermal expansion could potentially influence porosity and tortuosity, experimental and theoretical evaluations (see Supplementary Tables S4 (a) and S4 (b)) confirmed that such effects are minimal (<2.2 % porosity change in extreme cases). Therefore, tortuosity values obtained from uncharged electrodes are considered representative. Additionally, as mentioned earlier, the van der Waals radius of the electrolyte anions follows the order $BETI^- > FSI^- > PF_6^- > BF_4^-$: however, this size difference does not appear to influence the behavior of the porous electrodes.

From these findings, it was determined that R_{sol} and R_{ion} are governed by the conductivity of the electrolyte and the tortuosity factor of the electrode. Furthermore, it was quantitatively confirmed that using a high-conductivity electrolyte effectively reduces battery resistance. A strong correlation was observed between the conductivity of the bulk electrolyte and that within the electrodes: however, whether the same correlation holds for newly developed high-conductivity electrolytes, solvents, or various mixed compositions remains unclear, necessitating further study. Conversely, the development of electrolytes exhibiting exceptionally high ionic conductivity arising interactions at the solid/liquid interface within electrode pores should be pursued even if the conductivity of the bulk electrolyte is relatively low. Although this study is primarily focused on ionic resistance, the contribution of charge-transfer resistance to lithium intercalation must also be considered to fully evaluate the low-temperature performance and that after durability testing. Future studies will focus on the effects of interfacial membrane formation on electrolyte physical properties (including the conductivity, degree of dissociation, viscosity, and transport number) and charge transfer resistance, which may help elucidate the link between electrolyte composition, SEI stability, and long-term performance. And in this study, SOC 0 % and 50 % were selected as representative points for evaluating structural (torsional) and operational (R_{ion}) properties. Although the inclusion of a more extensive SOC analysis would add complexity due to compositional and morphological changes, a broader SOC-dependent analysis would be valuable and expected for future research.

4. Conclusion

A cylindrical battery was fabricated to evaluate the performance of four types of electrolytes with different characteristics. The initial performance of the batteries, measured in terms of direct current resistance and power capability, was compared and evaluated, and a cycle test was conducted at elevated temperatures to assess their practicality. Additionally, electrochemical impedance spectroscopy using a symmetric cell was performed to measure the various resistance components of the porous electrode. The relationship between electrolyte conductivity and the ionic resistance of the electrode was examined over a wide temperature range, and its effect on battery resistance was analyzed. The key findings are as follows:

1) Power Capability & Cycling Performance: Batteries using LiFSI exhibited the highest power capability, followed by $LiPF_6$ and LiBETI, while those using $LiBF_4$ had the lowest. Additionally, batteries with LiFSI and $LiPF_6$ demonstrated excellent cycling performance at elevated temperatures.
2) Electrochemical Resistance: Symmetric cells with LiFSI showed the lowest R_{sol} and R_{ion}, followed by $LiPF_6$ and LiBETI, whereas $LiBF_4$ had the highest resistances. These resistance trends quantitatively correlated with the power capability of the cylindrical battery.
3) Influence of Electrolyte Conductivity and Tortuosity: R_{sol} and R_{ion} were dependent on electrolyte conductivity and the tortuosity factor of the electrode. Additionally, it was quantitatively determined that using an electrolyte with high conductivity effectively reduces battery resistance.

Although this study particularly focused on R_{sol} and R_{ion}, the contribution of R_{ct} to lithium intercalation must also be considered to comprehensively evaluate the performance of the cells in low-temperature environments and following durability tests. Future studies will examine the relationship between the physical properties of the electrolyte and charge-transfer resistance during lithium intercalation.

CRediT authorship contribution statement

Yuichi Itou: Writing – original draft, Visualization, Supervision, Investigation, Formal analysis, Data curation, Conceptualization. **Nobuhiro Ogihara:** Writing – review & editing, Visualization, Project administration, Investigation, Formal analysis, Data curation, Conceptualization. **Shigehiro Kawauchi:** Writing – review & editing, Investigation, Formal analysis, Data curation.

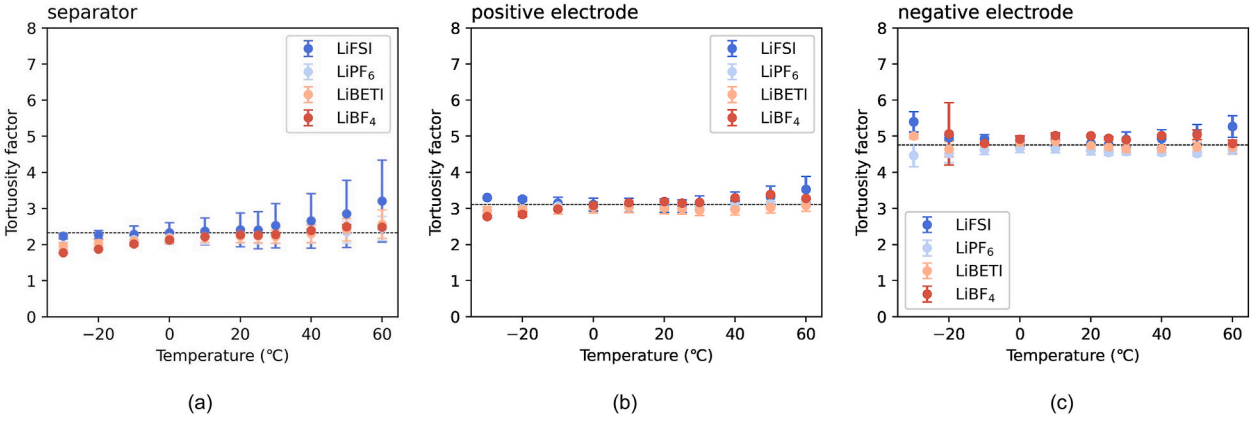


Fig. 7. Relationship between temperature and the tortuosity factor of the (a) separator, (b) positive electrode, and (c) negative electrode.

Supporting Information

This file provides additional electrochemical characterization and analysis data, including electrolyte injection volumes, cycling performance, resistance evolution, equivalent circuit models and fitting results, literature on SEI/CEI formation, and estimated porosity changes of electrodes under thermal and swelling conditions.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used [Chat GPT 4o] in order to [proofread the manuscript in the stage of revision]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpowsour.2025.238103.

Glossary

L	unit pore length, cm
L	characteristic distance of lithium-ion movement in the bulk electrolyte, cm
r	pore radius, cm
ρ	electrolyte resistance, cm
S	electrode surface area, cm^2
N	number of pores per unit electrode surface area
$Z_{i,\,L}$	interfacial impedance per unit pore length, Ωcm
$Z_{i,\,A}$	interfacial impedance per unit surface area, Ωcm^2
R_e	electric bulk resistance, Ω
R_{sol}	electrolyte resistance between two electrodes ($R_{sol} = \rho l/S$)
$R_{ion,\,L}$	ionic resistance per unit pore length ($R_{ion,\,L} = \rho/\pi r^2$), Ωcm^{-1}
R_{ion}	ionic resistance in pores ($R_{ion} = R_{ion,\,L}\,L/n$), Ω
$R_{ct,\,A}$	charge-transfer resistance per unit surface area, Ωcm^2
R_{ct}	total charge-transfer resistance ($R_{ct} = R_{ct,\,A}/2\pi rL$), Ω
$C_{dl,\,A}$	electric double layer capacitance per unit surface area, Fcm^2
C_{dl}	total electric double layer capacitance ($C_{dl} = C_{dl,\,A}\,2\pi rL$), F
C	the ratio of the charge/discharge current value to the battery capacity

Data availability

Data will be made available on request.

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